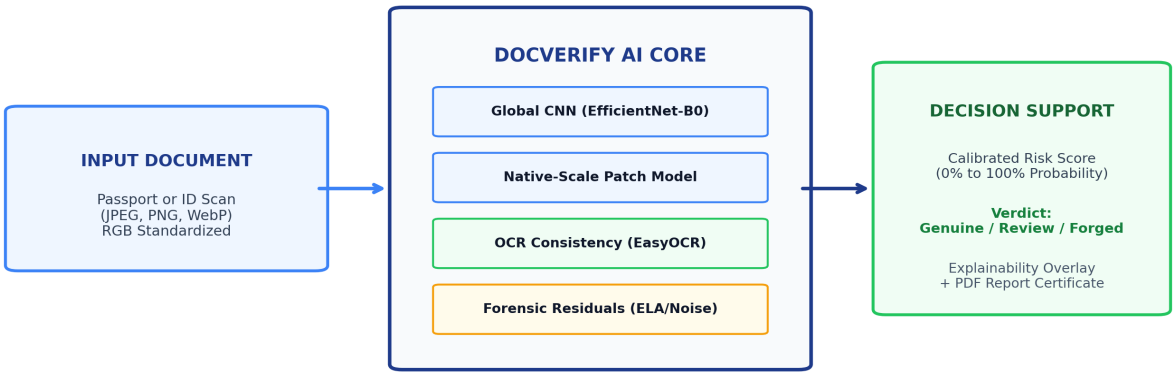
DocVerify — Phase 2: Global Baseline Model Training

Official Technical Milestone Report & Engineering Specification (Phase 2)

1. Phase Executive Summary & Objectives

Phase 2 trained the primary global classifier on full-frame document images. EfficientNet-B0 was selected for its high parameter efficiency (4.01M parameters) and compound scaling. The resulting model established the permanent global baseline (models/phase2/best_model.pth) achieving 79.69% accuracy on the test set.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Model Architecture	EfficientNet-B0 Binary Classifier (4.01M params)	Compact backbone enabling fast CPU/GPU inference with strong ImageNet feature transfer.
Accuracy Achieved	79.69% (255 / 320 correct on test set)	Baseline proof that global macro-texture contains significant forgery information.
Precision / Recall	Precision: 82.91% | Recall: 77.51% (F1: 80.12%)	Identified 131 true forgeries, but suffered 38 false negatives (missed forgeries).
ROC-AUC Score	0.9112 Area Under Curve	Demonstrated high overall separability between global authentic and fake features.

4. Key Discoveries, Challenges & Next-Phase Directives

- Global CNN effectively captured full-image macro features: paper texture, background Guilloche lines, and border artifacts.
- Suffered from False Negatives when tampering was restricted to tiny localized alphanumeric fields.
- Formally preserved best_model.pth as the frozen global backbone for all subsequent multi-modal phases.

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